IoT Smart Irrigation and Plant Monitoring System using ESP32

Submitted By

VIGNESHWAR ANDE

Department of Electronics and Telecommunication Engineering (ENTC)

Sandip University

Academic Year: 3/5

CERTIFICATE

This is to certify that the project entitled **"IoT Smart Irrigation and Plant Monitoring System using ESP32"** has been successfully completed by **VIGNESHWAR ANDE** in partial fulfillment of the requirements for the degree program in Electronics and Telecommunication Engineering at Sandip University.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Department of Electronics and Telecommunication Engineering, Sandip University, for providing the opportunity to complete this project.

I thank my faculty members, mentors, and friends for their valuable guidance and support throughout the development of this project.

ABSTRACT

Agriculture is one of the most important sectors contributing to economic growth. Traditional irrigation systems often result in water wastage and require continuous manual supervision. The objective of this project is to develop an IoT-based Smart Irrigation and Plant Monitoring System using ESP32.

The system monitors soil moisture, temperature, and humidity using sensors and displays real-time data on an OLED display. Based on soil moisture levels, a relay-controlled water pump can automatically irrigate plants. The project helps optimize water usage, reduce manual effort, and improve agricultural efficiency.

Keywords: ESP32, IoT, Smart Irrigation, Soil Moisture Sensor, DHT11, OLED Display, Relay Module.

1. INTRODUCTION

The Internet of Things (IoT) has transformed modern agriculture by enabling real-time monitoring and automated decision-making. Smart irrigation systems help farmers optimize water usage by providing precise irrigation based on soil conditions.

This project integrates an ESP32 microcontroller with a soil moisture sensor, DHT11 sensor, OLED display, relay module, and water pump. The system continuously monitors environmental conditions and can automate irrigation whenever soil moisture drops below a predefined threshold.

2. PROBLEM STATEMENT

Traditional irrigation systems face several challenges:

- Excessive water consumption
- Manual monitoring requirements
- Lack of real-time environmental data
- Inefficient irrigation scheduling
- Increased labor costs

The proposed system addresses these issues through sensor-based monitoring and automated irrigation.

3. OBJECTIVES

1. Monitor soil moisture levels.
2. Measure temperature and humidity.
3. Display sensor values on OLED display.
4. Control water pump automatically using a relay.
5. Reduce water wastage.
6. Improve irrigation efficiency.

4. LITERATURE SURVEY

Various IoT-based irrigation systems have been developed using microcontrollers such as Arduino, NodeMCU, and ESP32.

Recent studies demonstrate that:

- Soil moisture sensors provide accurate irrigation control.
- Wireless monitoring improves farm management.
- Automated irrigation reduces water consumption.
- ESP32 is widely used because of built-in Wi-Fi and processing capability.

The proposed system adopts a low-cost and scalable approach suitable for small-scale agricultural applications.

---

5. SYSTEM ARCHITECTURE

The system consists of:

Input Layer:

- DHT11 Sensor
- Soil Moisture Sensor

Processing Layer:

- ESP32 Microcontroller

Output Layer:

- OLED Display
- Relay Module
- Water Pump

System Flow:

Sensors → ESP32 → OLED Display

ESP32 → Relay → Water Pump

---

6. HARDWARE REQUIREMENTS

1. ESP32 (38 Pin)
2. DHT11 Sensor
3. Soil Moisture Sensor
4. OLED Display (128×64 I2C)
5. Relay Module (5V Active LOW)
6. Mini Water Pump
7. Breadboard

8. Jumper Wires
9. USB Cable

---

7. SOFTWARE REQUIREMENTS

- Arduino IDE
- ESP32 Board Package
- Adafruit SSD1306 Library
- Adafruit GFX Library
- DHT Sensor Library

Programming Language:

- Embedded C/C++

---

8. CIRCUIT CONNECTIONS

DHT11:

VCC → 3.3V

GND → GND

DATA → GPIO4

Soil Sensor:

VCC → 3.3V

GND → GND

AO → GPIO34

OLED:

VCC → 3.3V

GND → GND

SDA → GPIO21

SCL → GPIO22

Relay:

VCC → VIN (5V)

GND → GND

IN → GPIO26

Pump:

Battery (+) → Relay COM

Relay NO → Pump (+)

Battery (-) → Pump (-)

---

9. WORKING PRINCIPLE

1. ESP32 initializes all sensors and OLED display.
2. Soil moisture values are read from GPIO34.
3. Temperature and humidity are read from DHT11.
4. Data is displayed on OLED.
5. Soil moisture percentage is calculated using calibration values.
6. If moisture is below threshold, relay activates pump.
7. If moisture is above threshold, relay deactivates pump.
8. Process repeats continuously.

---

10. ALGORITHM

Step 1: Start System

Step 2: Initialize Sensors

Step 3: Read DHT11 Data

Step 4: Read Soil Moisture

Step 5: Calculate Moisture Percentage

Step 6: Update OLED Display

Step 7: Compare Moisture with Threshold

Step 8: Control Relay

Step 9: Repeat

---

11. RESULTS

Observed Outputs:

Temperature ≈ 32°C

Humidity ≈ 47%

Soil Moisture ≈ 88%

OLED Display Working Successfully

Relay Control Logic Working Successfully

The system successfully monitored environmental conditions and displayed real-time sensor data.

---

12. ADVANTAGES

- Water Conservation
- Low Cost
- Easy Installation
- Automatic Operation
- Real-Time Monitoring
- Low Maintenance

---

13. APPLICATIONS

- Smart Agriculture
- Home Gardening
- Greenhouse Monitoring
- Indoor Plant Care
- Research Projects

---

14. FUTURE SCOPE

• Wi-Fi Monitoring
• Mobile App Integration
• Cloud Database Storage
• AI-Based Irrigation Prediction
• Solar Powered System
• Weather Forecast Integration

---

15. CONCLUSION

The IoT Smart Irrigation and Plant Monitoring System demonstrates an efficient and low-cost solution for modern agriculture. By integrating soil moisture sensing, environmental monitoring, and automated irrigation, the system helps optimize water usage and improve crop management.

The project serves as a practical example of IoT implementation in smart farming and can be further expanded with cloud connectivity and advanced analytics.

REFERENCES

1. ESP32 Technical Documentation
2. Arduino IDE Documentation
3. DHT11 Sensor Datasheet
4. Soil Moisture Sensor Datasheet
5. Adafruit SSD1306 Library Documentation
6. IoT Smart Agriculture Research Papers